MIL-C-60899(MU)
8 October 1969
SUPERSEDING
FA-PD-MI-2440, Rev. 0
7 March 1961

MILITARY SPECIFICATION

CARTRIDGE, DELAY: M84, ASSEMBLY

1. SCOPE

1.1 This specification covers Cartridge, Delay: M84, Assembly, for use with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification For
MIL-C-60539	- Coating, Anodic, Conventional for Aluminum and Aluminum Alloys

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O-Rings
MIL-STD-1167	- Ammunition Data Card
MS-28741	- Hose Assembly, Detachable End-Fitting, Medium Pressure

DRAWINGS

Munitions Command

D10520085	- Cartridge, Delay; M84, Assembly

FSC 1377

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:05Z
Check the source to verify that this is the current version before use.

DRAWINGS (Cont'd)

C8864168	- Carton, Inner Packing, Ammunition For Cartridge Initiator, Delay M84, M85, M86, M87, M88, M89, M165, M128, M148, XM159, M174 or M175
C8864169	- Carton, Outer, Packing, Ammunition For Cartridge, Initiator, Delay, M84, M85, M86, M87, M88, M89, M165, XM128, XM148, M159, M174 or M175
C9214156	- Box, Packing, Ammunition For Cartridge, Initiator, Delay, M84, M85, M86, M87, M88, M89, M165, XM128, XM148, XM159, M174 or M175 (For level "A" packing only)
IEL10520085	- Inspection Equipment List for Cartridge, Initiator, Delay: M84, Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bid or request for proposal shall apply.

ASA B46.1 Surface Texture

(Application for copies should be addressed to the American Standards Association, 10 East 40th Street, New York, N.Y. 10016)

3. REQUIREMENTS

3.1 General. Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Protective finish. Protective finish shall be in accordance with MIL-C-60539.

3.3 Vibration. The cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the delay charge, primer, igniter or propellant or derangement of components.

3.4 Primer sensitivity. The primer shall fire when drop tested at an energy level of 26 inch ounces in a sectioned cartridge case, void of delay charge, igniter and propellant charge.

3.5 Functional.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:05Z
Check the source to verify that this is the current version before use.

3.5.1 The cartridge shall not fail to fire.

3.5.2 Ballistics. The cartridge when tested in an M14 or M15 Initiator, shall comply with the requirements prescribed in Table I.

TABLE I

Temperature	Pressure (psi) 1/ Minimum	Pressure (psi) 1/ Maximum	Delay Time, Seconds (see 6.2)
-65° to -70°F	1300	2200	4.2 max.
65° to 75°F	1500	2200	3.0 ± 0.3
160° to 165°F	1500	2200	2.3 min.

1/ Pressure values shall be produced in a one cubic inch pressure station at the end of 3 feet of hose (Size No. 4) in accordance with MS 28741.

3.6 Workmanship. Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects. The metal part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.6.2 Burr. The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 Foreign matter. The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facility suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection. Inspection shall be in accordance with MIL-A-2550.

4.2 First Article Sample.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:05Z
Check the source to verify that this is the current version before use.

4.2.1 Initial production sample. At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 23 Retaining rings and 23 Cartridge cases for protective finish tests, 50 cartridges for primer sensitivity tests, 50 for hardness tests and 30 cartridges for ballistic tests as indicated in Table II, 5 each for sealant application and 10 sets of unassembled components (including explosive components) for dimensional and visual examination.

TABLE II

Paragraphs Requirement	Test	Test Sample Size	Type of Test Destr.	Non-Destr.
3.2	4.4.1	Protective finish:		
		23 Retainers	X	-
		23 Cartridge Cases	X	-
3.3	4.4.2	Vibration (12 from ballistic sample)	-	X
Dwg.	4.4.3	Airtightness each		X
3.4	4.4.4	Primer sensitivity (50 each)	X	-
Dwg.	4.4.5	Hardness (50 each)	X	-
3.5.2	4.4.6.1	Ballistics (30 each)	X	-
Dwg.	4.4.7	Sealant Application (5 each)	X	-

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and tests. After examination and provisional acceptance at source, the sample shall be inspected for all requirements of the drawing and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics. The sample for ballistic tests shall be temperature conditioned and tested to determine compliance with Table I. The sample for firing shall consist of the following:

TABLE III

Temperature	Vibrated	Not Vibrated
-65° to -70°F	4	6
65° to 75°F	4	6
160° to 165°F	4	6

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:05Z
Check the source to verify that this is the current version before use.

4.2.1.3 Initial production sample failure. Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product. The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot. Unless otherwise specified by the procuring agency, an ammunition lot shall consist of 1200 cartridges, maximum, exclusive of test samples and shall consist of:

a. Components produced by a homogeneous process.
b. Delay elements from not more than one lot.
c. Primers from not more than one lot.
d. Igniters from not more than one lot.
e. Propellant from not more than one lot.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major or minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables IV and V. For an individual defect (single characteristic) the criteria in Table IV shall be used. For total defects within a major or minor class, the criteria in Table V shall be used. All non-conforming material shall be rejected. The roughness comparison specimens prescribed in ASA B46.1 shall be used as a basis of comparison for surface roughness determination. Inspection for rubber "O" rings shall be in accordance with MIL-STD-413. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter.

TABLE IV

Lot Size	Sample Size	Major		Minor	
		Accept	Reject	Accept	Reject
1-300	100% examination	-	-	-	-
301-1300	110	1	2	2	3
1301-3200	150	2	3	3	4
3201-8000	225	3	4	4	5
8001-22,000	300	4	5	5	6

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:05Z
Check the source to verify that this is the current version before use.

TABLE V

Sample Size for Major Defects Number of Defects in Class	Total Allowable Defects Per Sample Size 110	150	225	300
2-3	1	2	3	4
4-7	2	3	4	5
8-19	3	4	5	7
20-43	4	5	8	10
over 43	6	8	11	14

Sample Size for Major Defects Number of Defects in Class	Total Allowable Defects Per Sample Size 110	150	225	300
2-3	2	3	4	5
4-7	3	4	5	7
8-16	4	5	8	10
17-48	6	8	11	14
over 48	8	11	17	20

4.3.2.1 Ring, Retaining, Dwg. C8597235, covering a detail of Dwg. D10520085.

DEFECT	METHOD OF INSPECTION
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Concentricity of inside diameter with outside diameter	Gage
102. Diameter inside, max.	Gage
103. Diameter, outside, min.	Gage
104. Protective finish inadequate or defective	Visual
105. Thickness	Gage
106. Metal defective	Visual
107. Burr	Visual
108. Foreign matter	Visual
MINOR	
201. Diameter, inside, min.	Gage
202. Diameter, outside, max.	Gage
203. Total length	Gage
204. Radius missing or incorrect	Visual
205. Surface finish improper	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:05Z
Check the source to verify that this is the current version before use.

4.3.2.2 Case, Cartridge, Drawing C10519996, covering a detail of Dwg. D10520085

DEFECT	METHOD OF INSPECTION
CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Concentricity of cartridge head with body	Gage
102. Straightness of base	Gage
103. Thickness of base	Gage
104. Thickness of wall	Gage
105. Protective finish inadequate or defective	Visual
106. Burr or sharp corner	Visual
107. Foreign matter	Visual
108. Diameter of head, min.	Gage
MINOR	
201. Diameter of head, max.	Gage
202. Diameter of body	Gage
203. Total length	Gage
204. Location of datum diameter	Gage
205. Thickness of wall, max.	Gage
206. Thickness of base, max.	Gage
207. Radius missing or incorrect	Visual
208. Surface finish improper	Visual

4.3.2.3 Delay Element, M25, Dwg. D10520091, covering a detail of Dwg. D10520085.

CRITICAL	
1. Metal defective	Visual
2. Mixed ammunition	Visual
3. Flash hole missing or obstructed	Visual
4. Marking missing, incorrect or illegible	Visual
5. Delay charge not visible through flash hole	Visual
MAJOR	
101. Protective finish inadequate or defective	Visual
102. Foreign matter	Visual
MINOR	
None defined	

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:05Z
Check the source to verify that this is the current version before use.

4.3.2.4 Cartridge, Delay, M84, Assembly, Dwg. D10520085

DEFECT	METHOD OF INSPECTION
CRITICAL	
1. Any part missing	X-ray 1/ and Visual
2. Improper assembly	X-ray 1/ and Visual
3. Mixed ammunition	Visual
4. Metal defective	Visual
5. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Concentricity of cartridge head with body	Gage
102. Diameter of cartridge head recess, min.	Gage
103. Diameter of body, max.	Gage
104. Diameter of cartridge head, max.	Gage
105. Depth of cartridge head recess	Gage
106. Location of datum diameter	Gage
107. Cartridge profile failure	Gage
108. Firing pin or retaining cup above flush with face of head	Gage
109. Foreign matter	Visual
MINOR	
201. Total length, max.	Gage
202. Varnish missing or inadequate	Visual
203. Fold, wrinkle, dent, buckle, bulge or scratch	Visual

1/ The assembled cartridge shall be x-rayed with the long axis of the cartridge perpendicular to the rays of the x-ray machine. The x-ray negatives, properly identified, shall be forwarded to the procuring activity.

4.3.2.5 Carton, Inner, Packing, Dwg. C8864168

CRITICAL

None Defined

MAJOR

None Defined

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:05Z
Check the source to verify that this is the current version before use.

4.3.2.5 (Cont'd)

DEFECT	METHOD OF INSPECTION
MINOR	
201. Contents loose	Manual and Visual
202. Quantity of contents incorrect	Visual
203. Separator or filler missing	Visual
204. Heat sealer incomplete, loose, or distorted	Visual
205. Cut or damaged through impregnated box	Visual
206. Marking missing, incorrect, or illegible	Visual

4.3.2.6 Carton, Outer Packing, Ammunition, Dwg. C8864169

CRITICAL

None defined

MAJOR

None defined

MINOR	
201. Contents loose	Manual and Visual
202. Cut or damaged through impregnated carton	Visual
203. Contents of carton exposed	Visual
204. Carton damaged	Visual
205. Marking missing, incorrect or illegible	Visual
206. Number of inner cartons incorrect	Visual

4.3.2.7 Box, Packing, Ammunition, Dwg. C9214156.

CRITICAL

None defined

MAJOR

None defined

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:05Z
Check the source to verify that this is the current version before use.

4.3.2.7 (Cont'd)

DEFECT	METHOD OF INSPECTION
MINOR	
201. Contents loose	Manual and Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of cartons incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.3.3 Inspection testing. The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Protective finish. Twenty-three retainers and 23 cartridge cases shall be selected from each lot for protective finish tests. Failure of any component to comply with the requirements of 3.2 shall be cause for rejection of the lot of components.

4.3.3.2 Vibration.- Twelve of the 30 assembled x-rayed cartridges of the initial production sample shall be subjected to vibration in accordance with 3.3. After vibration the 12 vibrated cartridges shall be x-rayed as directed in note 1/ 4.3.2.4. Failure of any cartridge to comply with the requirements of 3.3 shall be cause for rejection of the lot. When there is a change in the method of charging the cartridge or in the lot of primers, 12 additional cartridges shall be selected and tested as indicated herein.

4.3.3.3 Airtightness. Each cartridge shall be subjected to an airtightness test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the cartridge.

4.3.3.4 Primer sensitivity. A sample of 50 assembled cartridges from the initial production sample and 25 assembled cartridges from each subsequent production lot shall be tested for sensitivity. Failure of one cartridge to comply with the requirements of 3.4 shall be cause for rejection of the lot.

4.3.3.5 Hardness. Fifty units from each heat treated lot shall be selected for hardness tests. Failure of any unit to comply with drawing requirements shall be cause for rejection of the lot of components.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:05Z
Check the source to verify that this is the current version before use.

4.3.3.6 Functional. The following tests shall be performed by the government.

4.3.3.6.1 Ballistics. Thirty test cartridges from each production lot shall be grouped and temperature conditioned in accordance with 4.4.6.1.1 and Table VI. Failure of any cartridge to comply with the requirements of 3.5.2 shall be cause for rejection of the lot.

4.3.3.7 Sealant application. Five cartridges from each production lot shall be selected for this test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the lot.

TABLE VI

Temperature	Sample Size
-65° to -70°F	10
65° to 75°F	10
160° to 165°F	10

4.3.4 Inspection equipment. The examination and tests shall be made using equipment listed on IEL 10520085.

4.4 Test methods and procedures.

4.4.1 Protective finish. Test procedures for protective finish shall be in accordance with MIL-C-60539. The retainers and cartridge cases shall be sectioned longitudinally to better expose the critical surfaces to the salt spray.

4.4.2 Vibration. The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.3 Airtightness. The method and procedure for performing the airtightness test shall be by use of equipment listed on IEL10520085.

4.4.4 Primer sensitivity. Remove and discard the charged delay body and all powder charges from the test cartridges by sectioning the cartridge case around its periphery, one inch, maximum, above the base. Drop test 50 sectioned cartridges from the initial production sample and 25 sectioned cartridges from each production lot (all void of charged delay body and powder charges) at 13 inches, using a steel ball 15/16 inches in diameter, weighing 1.94 ounces plus or minus 0.02 ounces, in a primer sensitivity machine with a cartridge holder approved by the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:05Z
Check the source to verify that this is the current version before use.

4.4.5 Hardness. A hardness testing machine shall be used to determine the hardness values specified on the applicable drawing.

4.4.6 Functional.

4.4.6.1 Ballistics. The sample for ballistic tests shall be grouped and temperature conditioned in accordance with 4.4.6.1.1 and rigidly supported in the test fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample unit shall be reconditioned in accordance with 4.4.6.1.1.

4.4.6.1.1 Temperature conditioning. A standard inert cartridge containing a temperature sensitive element shall be placed in the conditioning chamber at the same time as the sample. The conditioning time shall be a minimum of one hour longer than the time necessary to bring the element contained in the dummy cartridge to the required temperature.

4.4.7 Sealant application. The cartridge shall be assembled in such a manner that determination can be made that the sealant has been applied for the full length (360°) of the joining surfaces at the junction of the "O"-ring and case sidewall and that the propellant cavity is free of sealant.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking. Packing and marking shall be in accordance with Drawing C8864168, C88644169 and C9214156.

5.2 Data card. Data card information shall be in accordance with MIL-STD-1167.

6. NOTES

6.1 Ordering data. Procurement documents should specify the title, number and date of this specification.

6.2 Delay time is defined as the time that elapses between the firing of the primer and the beginning of a continuous pressure rise on a pressure time record.

6.3 Supersession date. This specification includes the requirements of purchase description FA-PD-MI-2440 dated 7 March 1961.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1377-A267

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:05Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:05Z
Check the source to verify that this is the current version before use.

FOLD

Department of the Army

Frankford Arsenal
Philadelphia, Pa. 19137

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
Department of the Army

Commanding Officer
Frankford Arsenal
ATTN: SMUFA-N1100
Philadelphia, Pa. 19137

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T01:05Z
Check the source to verify that this is the current version before use.